



Computer Networking Fundamentals



Computer Network Fundamentals

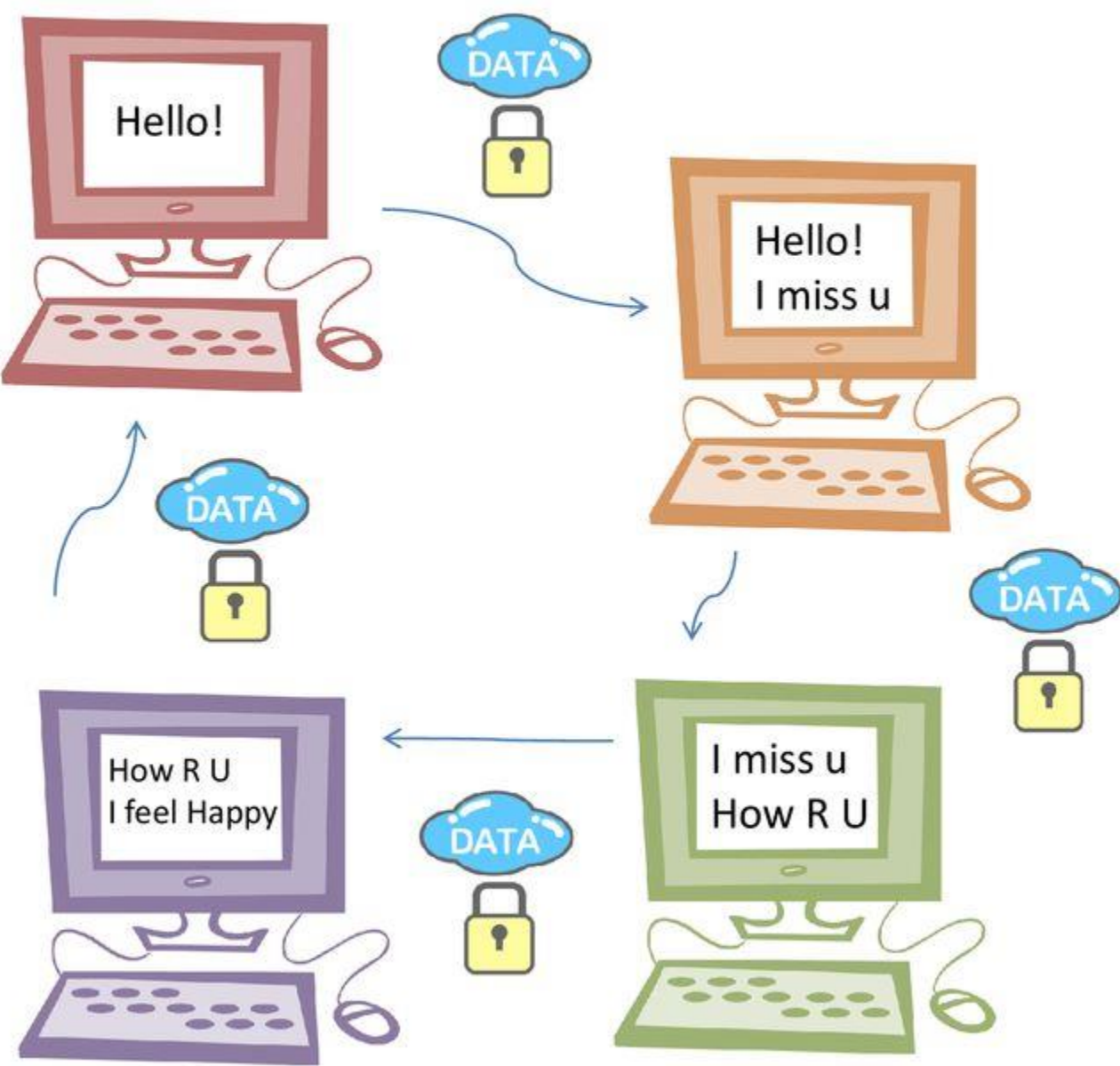
Lesson Objectives



- Describe a Computer Network
- Identify the advantages and Disadvantages of a computer network
- Identify the types of computer network
- Identify the physical components of a computer network

Activity 1

Describe this picture??



2 Minutes



Two or more computers that exchange data



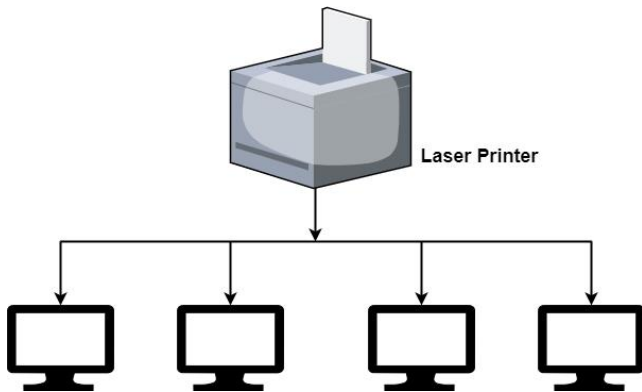
A network is a set of technologies- including hardware, software, and media- that can be used to connect computer together, enabling them to communicate, exchange information, and share resources in real time.

What are the Uses of Computer Network?

Computer Network provides several uses, both to organizations and individual users.

1. Resource Sharing

It defines creating all programs, peripherals and data feasible to anyone computer on the network to all other computers without considering the physical areas.

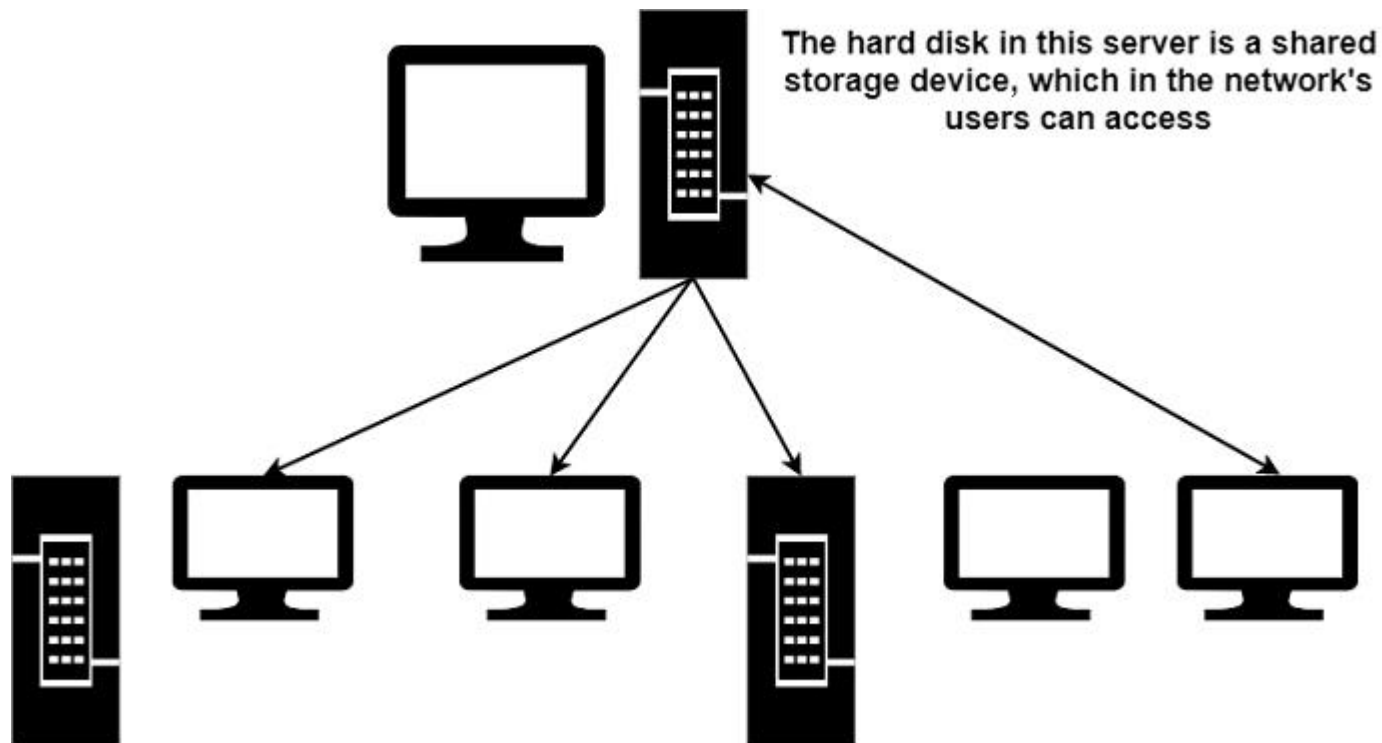


Another method of resource sharing is load sharing. If a job is needed, it can be implemented using various computers in a network by partitioning it, which reduces time consumption and loads both things for a particular computer.

2. Simultaneous Access

computer network allows few users to create programs and data at the equivalent time. An example is a company's quarterly sales document, which several managers need to view and update. We can store the information on a network server, which is a central computer with a huge storage device and other resources that all users can share.

If the server stores files for users to access, it is generally known as a file server. We can save an individual copy of a data file on the server, accessible to each employee in the organizations, as display in the figure. Moreover, if one user changes the file, other users will see the difference when using it.

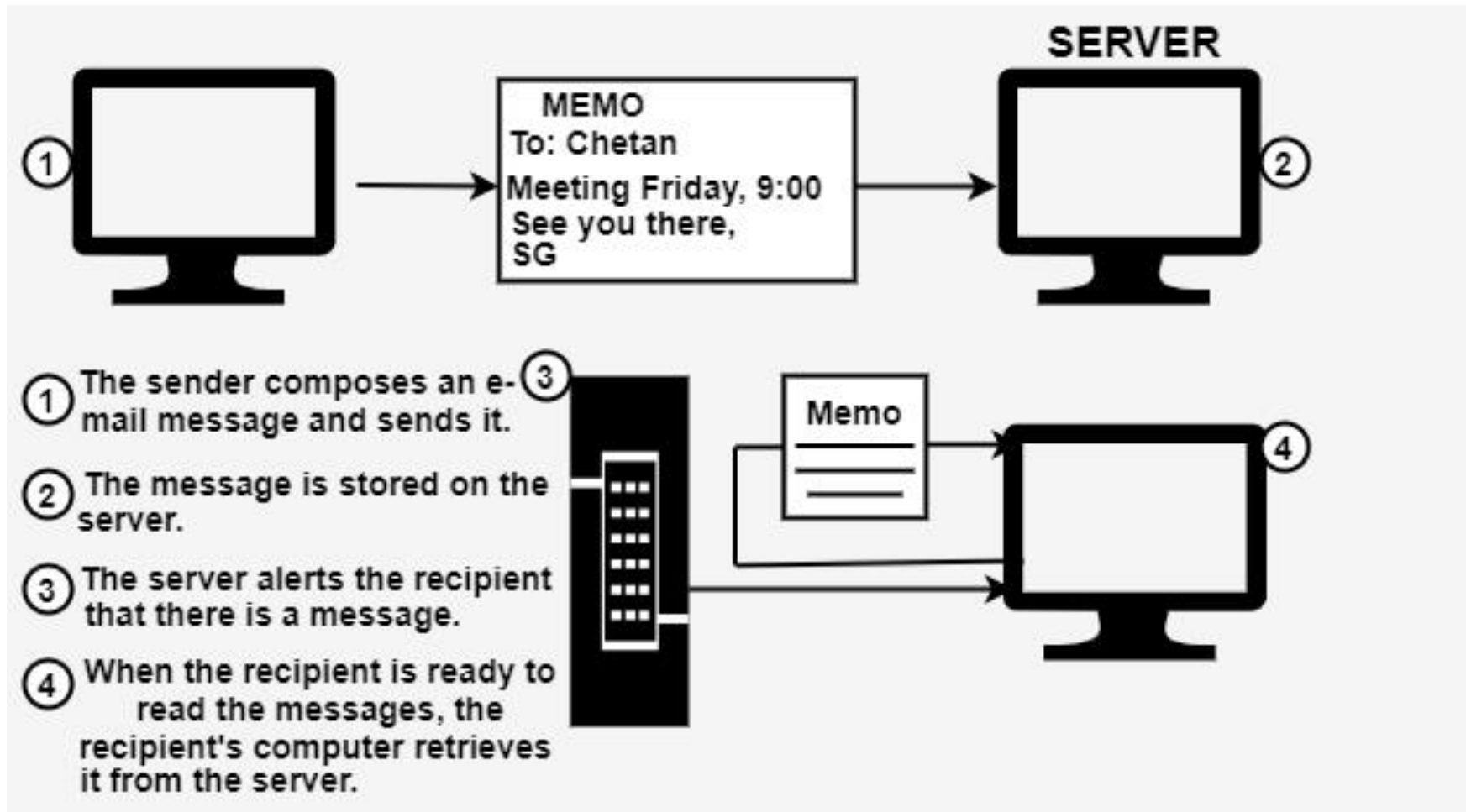


3. Person-to-Person Communication

Person to person communication involves the following –

- **Electronic-mail (e-mail)**

Sending and receiving email over a typical network is shown in the figure –



Others usage:

- **Teleconferencing and Video Conferencing**

A teleconference is a virtual conference in which various people in multiple areas conduct discussions by typing messages. All the other people can see each message in the teleconference.

If clients have the required hardware and software, they can view and talk to each other as they face online instead of merely typing messages. This process is known as videoconferencing.

- **Worldwide Newsgroups**

Worldwide newsgroups in which one individual posts information and few other subscribers can interpret it or provide feedback.

- **Interactive Entertainment**

Interactive entertainment includes the following –

- **Video on Demand** – It is feasible to choose any movie and displayed it on your screen.

- **Game Playing** – You can enjoy multiperson real-time simulation games. It can participate in live TV programmers like quiz, contest, discussion, etc.

Activity 2

Sort out the advantages and disadvantages of computer network??



2 Minutes



Advantages



Disadvantages



Advantages



Sharing Files, data & media

Ease of communication

Centralized data

Save money

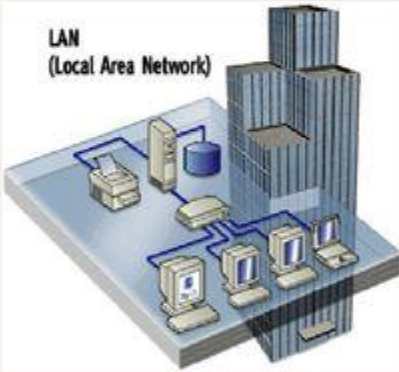
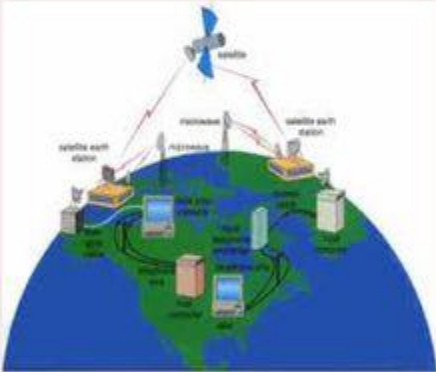
Disadvantages

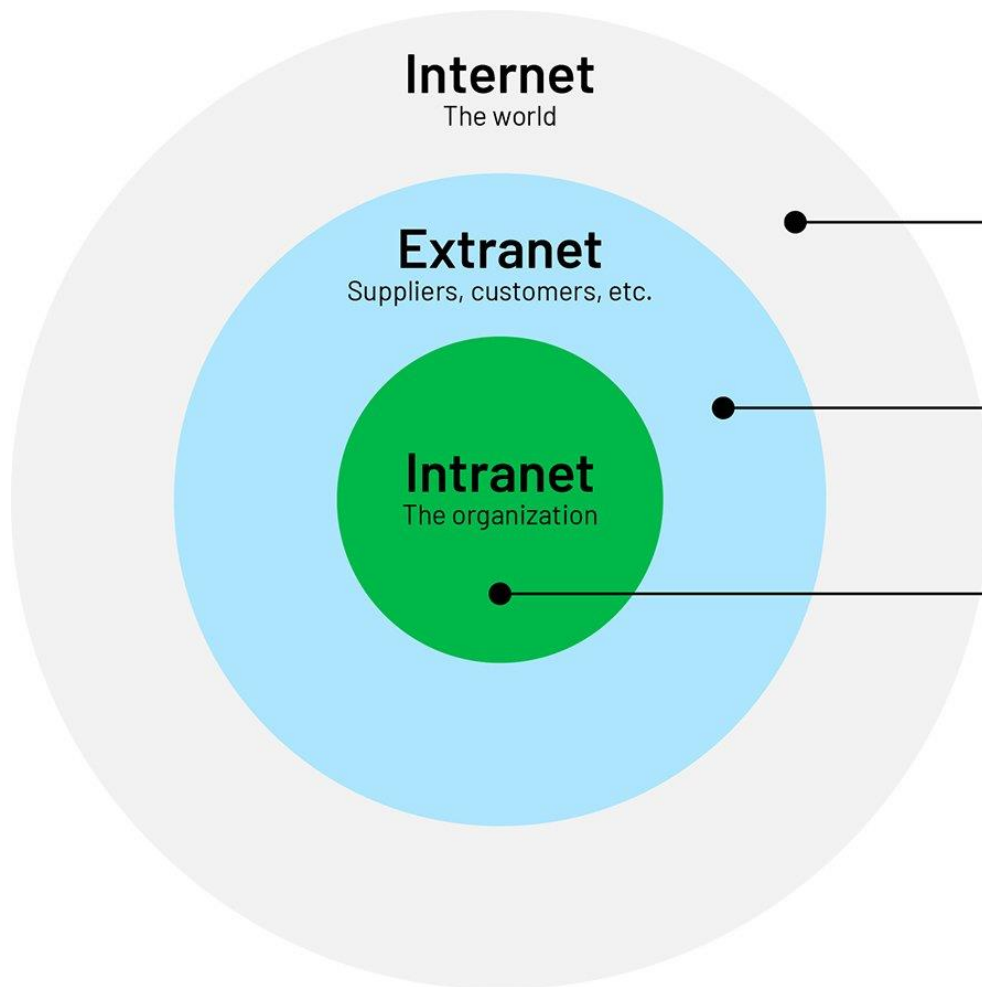


Easy to detect a virus

Less Security on devices and files

Types of Computer Network (Geographical Area)

Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
PAN	LAN	MAN	WAN
			
Computer Network Built around for individual use including PCs, Printers, Phones..etc	Covers small geographical area such as a single building, school	Covers a small city as a university campus and office buildings.	Covers a wide area as a whole country or world.



Internet

The world

Extranet

Suppliers, customers, etc.

Intranet

The organization

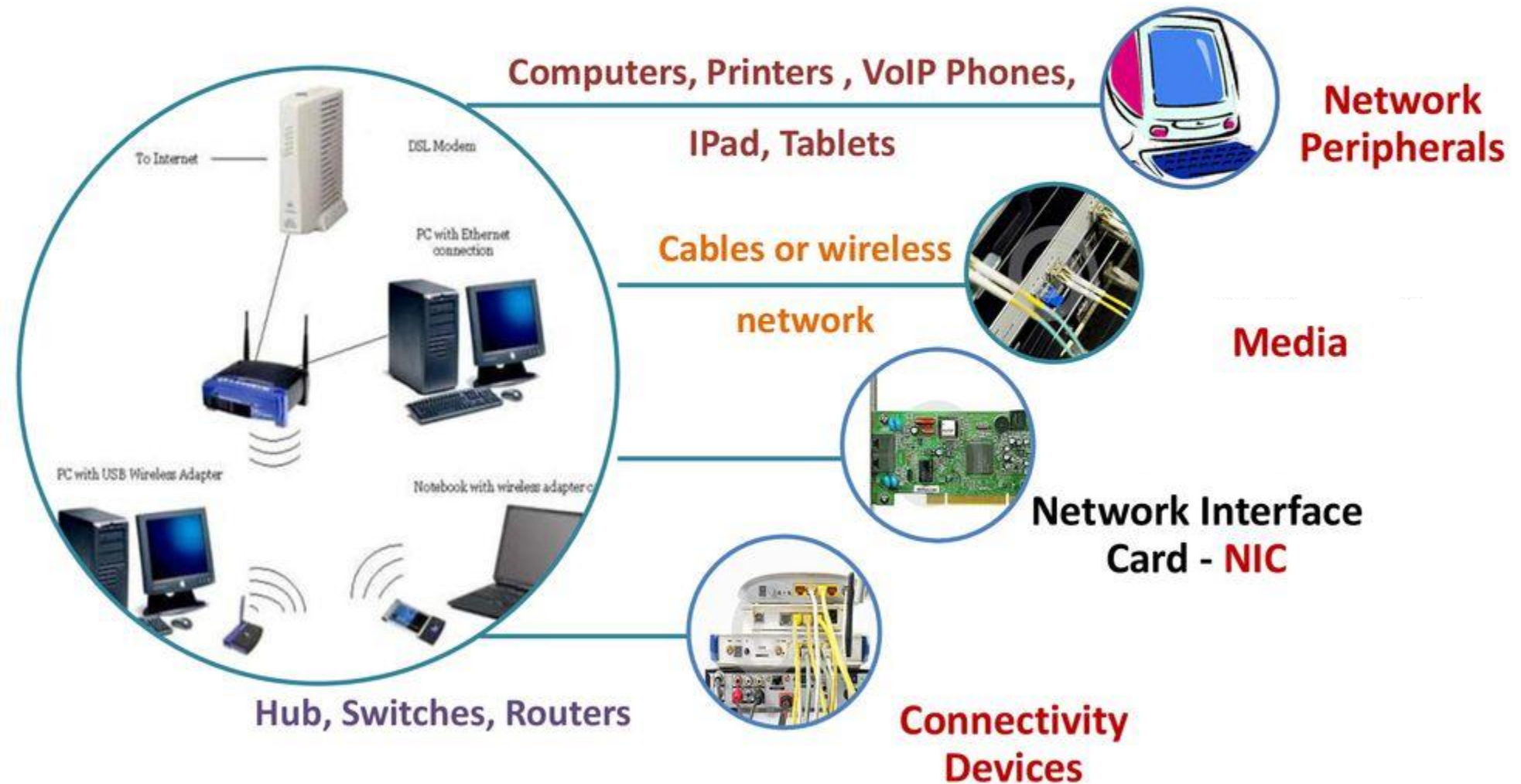
The **internet** creates connections between computers around the world.

An **extranet** creates connections beyond (or outside) an organization.

An **intranet** creates connections inside an organization.

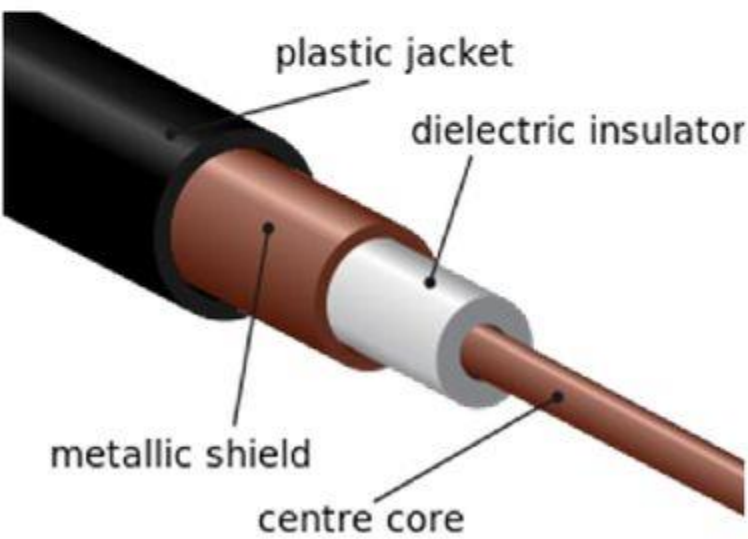
Parameter	Intranet	Extranet
Usage	Private	Private
User Types	Organization employees and Internal company departments	Suppliers, customer and Business partners.
Usage	Internal employee communication , telephone directories etc.	Check status of orders, Access data , send email
Security	High security. Configured under 100 security level in firewall	Generally uses VPN technology for secured communication over Internet. Medium security Level.
Regulated by	It is regulated by an organization.	It is regulated by multiple organization.
Ownership	Owned by Single organization	It is owned by single/multiple organization.

Networks Hardware



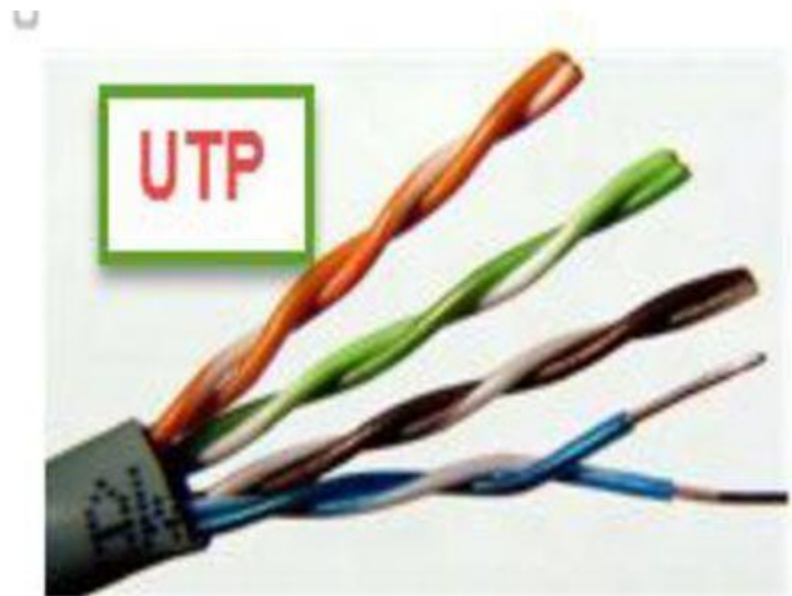
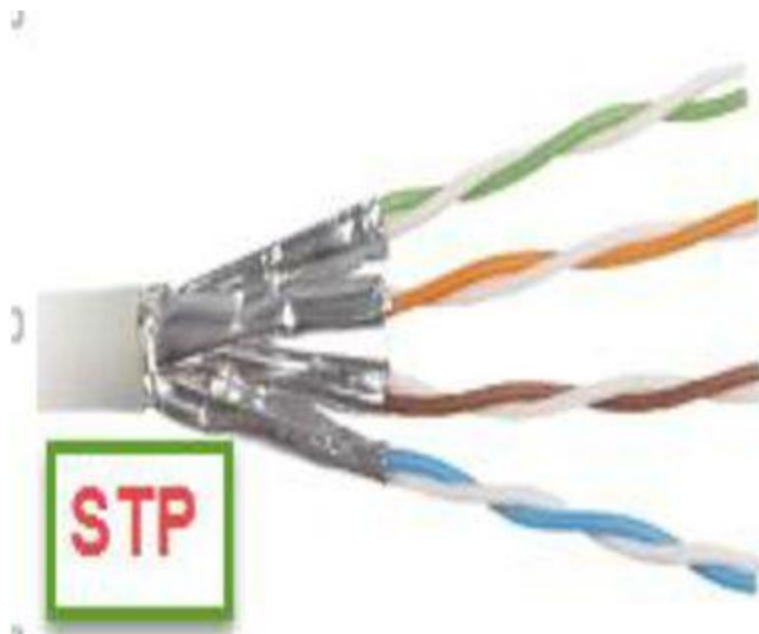
Coaxial Cable(COAX)

Coaxial Cable (COAX): It's a fairly common type of shielded data transmission cable, which is made up of two conductors that are coaxially oriented but separated by a layer of insulation. Data transfer speed has a rate between (10 to 100 Mbps), Maximum cable length 500 meters.



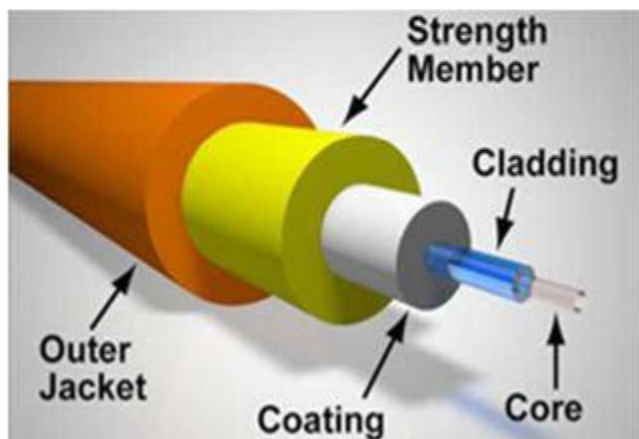
Twisted-Pair Cable

Twisted-Pair Cable: Twisted-pair cables can be either STP (shielded twisted pair) and UTP (unshielded twisted pair), data transfer speed has a rate between (10 to 1000 Mb / s), and maximum cable length 100 meters.



Fiber-Optic Cable

Fiber-Optic Cable: is made up of 100 or more incredibly thin strands of glass or plastic known as optical fibers. each of which is capable of transmitting messages modulated onto light waves. data transfer speed has a rate between (100 Mb/s to 1 Gb/s), and is used to data transfer for very long distances.

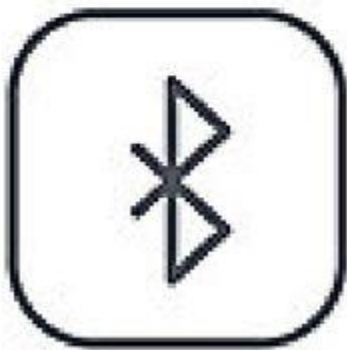


For more information about network , visit the website
<http://www.homepcnetwork.com>

WIRELESS NETWORKS

Wireless: In some types of networks may not prefer or cannot extend the cables and thus media can be used for wireless sending and receiving data.

- Radio Frequency (RF) like Wi-Fi, 3G, 4G and Bluetooth
- Microwave
- Satellite
- Infrared



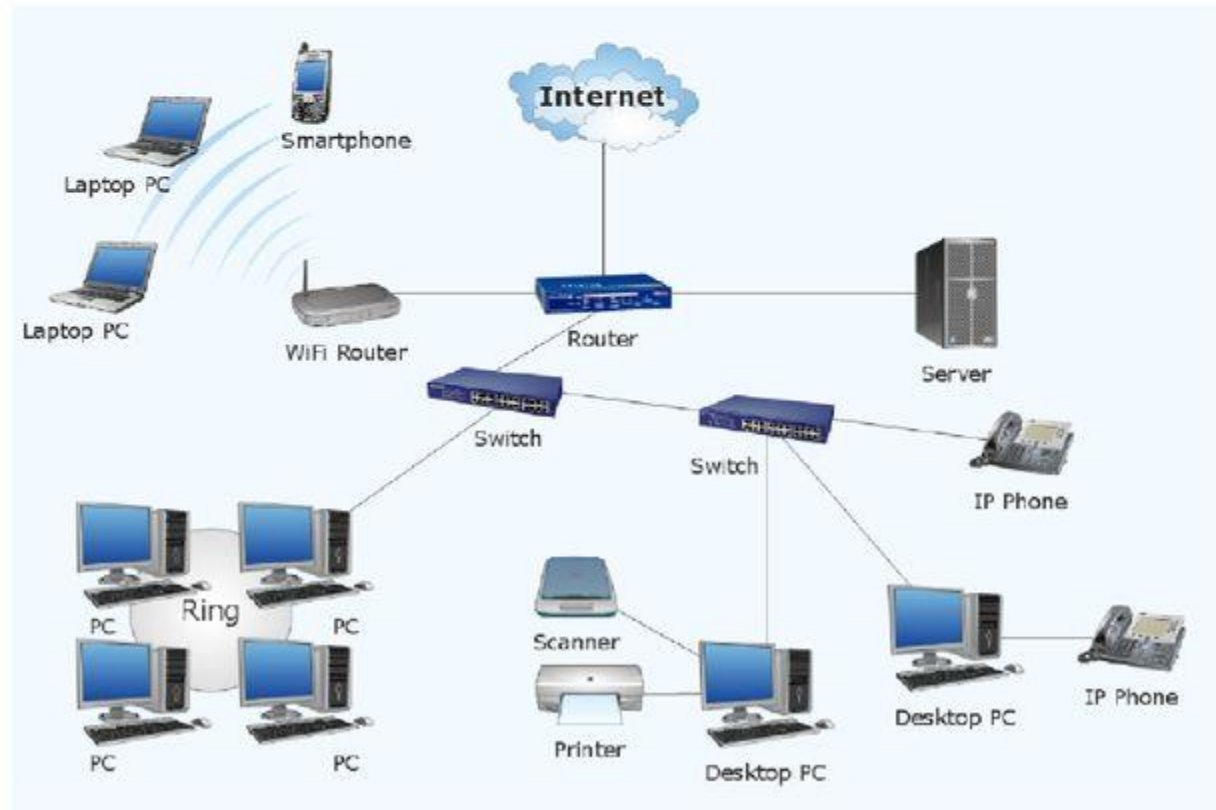
Network Interface Card (NIC):

3- Network Interface Card (NIC): is the hardware device that enables you to send and receive data to and from your computer. An adapter can connect to the network by cable (wired) or by air (wireless)



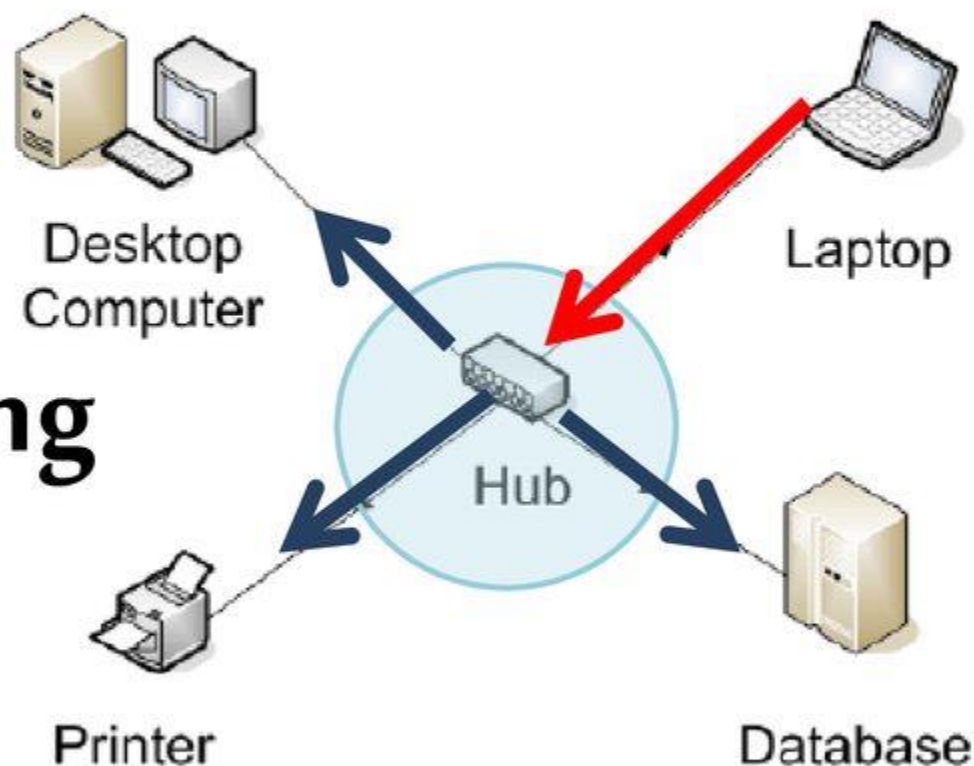
Networks Hardware – Connectivity Devices

Connectivity Devices: Are devices that work on connecting peripheral devices using common media of data transfer in order to transfer data between these devices as well as other functions related to improving network performance



1. Hub

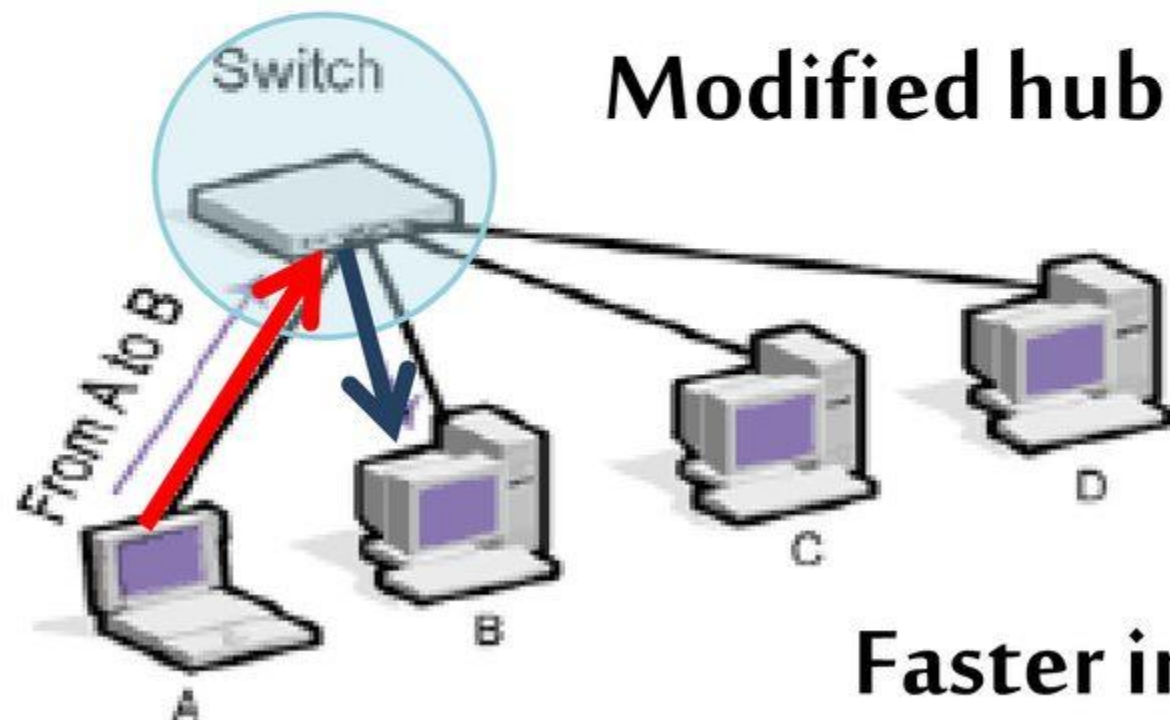
**Central connecting
device**



Is the network will be fast??

2. Switch

Switch: Switches work the same way as a hub, but they can identify the intended recipient of the data. Switches can send and receive data at the same time.



Router

Connect between two networks

